

Step 4 One-Pager — Game Abstraction and Scaling

Research on the possibilities for applying Artificial Intelligence in computer games

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Problem. Step 3 established that a solver’s reach is bounded by tree size, so the classical route to real poker is not a better solver but a *smaller game*: shrink it, solve the shrunken version, translate the strategy back, and pay an exploitability tax for the distinctions you threw away. Step 4 measures that tax. The question is not “does abstraction help” but which axis of compression is safe, and whether the compute a reduction buys ever repays the strategic information it destroys.

Approach. A Leduc-family pipeline that compresses along three axes and evaluates every resulting strategy **in its own full game**: **lossless suit isomorphism** (Leduc payoffs depend only on rank and on whether the private card pairs the board, so suit-isomorphic information sets merge with no loss), **lossy card bucketing** (hand-strength features, EMD-style distances, k-means, in perfect- and imperfect-recall regimes), and **action abstraction plus translation** (nearest-action, probability-split, pseudo-harmonic) on a variable-bet Mini-NL Leduc. Benchmarked with CFR+ under a common **180-second** budget, 3 seeds, on fixed-limit Leduc, Mini-NL Leduc and a 4-rank Extended Leduc. **All numbers below are measured.**

Key results (measured).

- *Lossless compression is the highest-value move, and it wins by buying iterations.* Suit isomorphism cuts fixed-limit Leduc from **936 to 288** information sets and reaches **3.13e-6** exploitability against full CFR+’s **4.44e-5** in the same wall clock — because it completes **14,185** iterations instead of **2,655**. Confirmed at scale on Extended Leduc: **10,304 -> 2,968** info sets (-71%), **0.0272 -> 0.00126**, roughly 6x the iterations.
- *Lossy card buckets install a floor that compute cannot lift.* **k2 0.571, k3 0.382, k5 0.382** — all at 11,000+ iterations, i.e. 4x more solving than the full game and four orders of magnitude worse. The error is in the abstraction, not the budget.
- *Action abstraction is the dangerous axis.* Mini-NL: the full 4,704-info-set game reaches **0.00692**, while the 936-info-set action abstraction reaches **0.673** despite 5x the iterations — roughly **100x worse** for a 5x smaller tree. Extended Leduc is starker still: suit+action **4.696** and suit+action+buckets **4.734**, both far worse than simply solving the unabridged game. Translation error dominates the total.
- *A measured artifact, not a law.* **k5** matches **k3** exactly because Leduc’s post-flop hand-strength distributions collapse into three effective shapes; the bucket-count ordering seen here is a property of this tiny game.
- *Stated limitations.* CFR+ is deterministic, so the three seeds are not independent stochastic runs; the action-abstraction numbers depend on the current translator and deployment semantics and should be read as a diagnostic failure mode; and the OpenSpiel comparison aligns all 936 information sets as a sanity check, not as proof of identical solver dynamics.

Thesis connection. The reporting format is the transferable part: strategy quality is meaningless without game size beside it, and the Pareto frontier of the two is an early seed of Contribution #3’s evaluation methodology. The mechanism transfers as well — Step 7’s type-based opponent modelling is this same “group for tractability” idea moved from state space to strategy space, and its confident-but-wrong failure is action translation’s misspecification in another costume.

Open questions. Static translation is brittle by construction, which points at subgame and nested solving (Step 6) as the production-grade answer to off-tree actions. Whether the ordering measured here — card abstraction survivable, action abstraction ruinous — holds in a game with enough hand-strength diversity that **k5** genuinely differs from **k3**. And whether an abstraction can be adapted to the opponent rather than fixed in advance, which is where this step touches the adaptive framework directly.